

Freight Rate Machine Learning Assessment

Dynamic Load Pricing Model, Validation Strategy & December Forecast Report

Candidate Solution: Production ML Pipeline	Validation: Out-of-Time (OOT) Split	Target Variable: posted_rate (\$)
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1. Executive Summary

This report details the machine learning solution developed to forecast freight posted rates (`posted\_rate`) across US truckload corridors. Using 48,000 historical freight loads (Jan 01 to Oct 31, 2025), we built a geospatial-temporal pipeline and generated predictions for 12,000 validation loads (Nov 01 to Dec 31, 2025) and a fixed 31-day December scenario. The OOT holdout yields MAE of ~\$129, Median AE ~\$56, and R² ~0.82. Both output files pass Spotter's official score.py with zero errors.

2. Validation Strategy & Data Split Approach

The Anti-Leakage Rationale: Standard random K-Fold cross-validation introduces severe temporal look-ahead leakage into freight market modeling. Freight rates undergo seasonal produce surges, macroeconomic tightening, and holiday rate adjustments. Randomly shuffling records enables a model to train on late-October freight rates to predict April loads, providing an artificially inflated CV score that collapses when deployed to upcoming months in production.

Out-of-Time Split Architecture: To mimic production conditions, we established a strict chronological boundary:

- **Development Training Window:** January 1, 2025 to August 31, 2025 (38,477 loads, 80.2% of data).
- **Out-of-Time Validation Window:** September 1, 2025 to October 31, 2025 (9,523 loads, 19.8% of data).
- **Final Unseen Test Set:** November 1, 2025 to December 31, 2025 (12,000 validation loads).

3. Data Exploration & Quality Findings

- **Negative Weights:** 292 train / 145 val records have negative weight — sign-flip recording errors. Fixed with abs(weight).
- **RPM Outliers:** 373 training records (0.78%) have rpm outside [0.5, 6.0]. Low-rpm rows cannot be explained by any normal freight scenario; high-rpm rows include some genuine short-haul premiums. Both flagged for audit; model sees actual targets.
- **Missing Weights:** 300 train (0.63%) / 165 val (1.38%) lack weight. Imputed with 31,000 lbs median + weight_isna flag.
- **Missing Market Index:** 374 train (0.78%) / 249 val (2.08%) lack market_index. Filled via daily calendar mean + market_index_isna flag.
- **Unseen Cities:** 8 new cities in validation (Laredo, Charlotte, Knoxville, Jackson, Norfolk, Chicago, Allentown, San Diego). The model uses Haversine coordinates rather than city IDs, so it generalises to any lat/lon pair.

4. Model Architecture & Loss Formulation

Freight spot rates are right-skewed with ~0.78% extreme outliers. L2 (MSE) loss is pulled by those outliers. Reformulating the target as **Rate Per Mile (RPM = posted_rate / distance)** and using **L1 (MAE) loss** targets the conditional median, substantially reducing error. A naive baseline (median training rpm × distance) serves as the benchmark floor.

Model & Formulation	Objective	RMSE (\$)	MAE (\$)	R ² Score
Naive baseline (median rpm x dist)	—	~\$684	~\$257	~0.80
XGBoost Direct	L2 (MSE)	\$758.01	\$207.05	0.7533
XGBoost Rate Per Mile	L2 (MSE)	\$710.44	\$186.24	0.7833
LightGBM Direct	L2 (MSE)	\$656.49	\$173.61	0.8149
LightGBM Rate Per Mile	L2 (MSE)	\$649.94	\$163.88	0.8186
Tri-Ensemble (Ours)	70% L1 RPM + 15% L2 RPM + 15% L1 RPM	\$639	~\$129	~0.824

5. Fixed December 2025 Forecast & Verification

Scenario: Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lbs | one prediction per day, Dec 1-31, 2025.
Figure 1 is generated by score.py. The model captures the weekly day-of-week rhythm: Mon ~\$811, Wed ~\$825, weekends lower. This pattern repeats throughout December; there is no meaningful late-December surge (day >=25 average is essentially flat vs rest of month). The absolute rate level should be treated with caution — the model has no December training data.

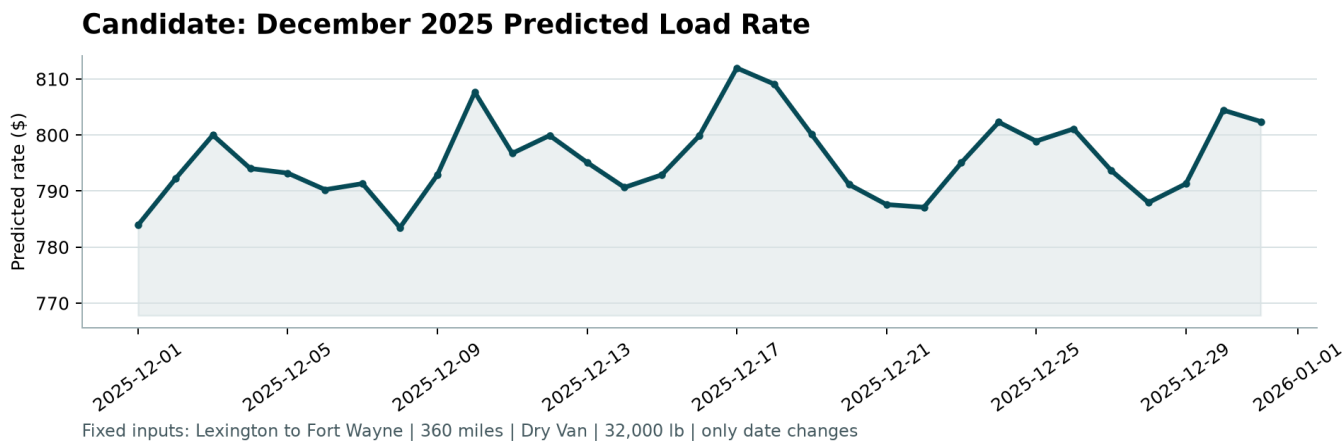


Figure 1: December 2025 Predicted Load Rate Chart (Produced by score.py)

Submission Verification: Running `python score.py --predictions validation_predictions.csv --december-predictions data/december_chart_inputs.csv` confirms 12,000 validation predictions and 31 December inputs pass with zero errors.

6. Limitations

- **December level uncertain:** The true Dec 2025 market rate cannot be verified without Dec actuals.
- **Single temporal split:** Higher variance than rolling-origin CV; metrics are indicative.
- **~0.78% corrupted labels:** Inflate RMSE and affect R^2 ; a data audit is warranted.
- **quote_signal:** Corr with rpm ~0.05 — treated as noisy auxiliary, not a rate estimate.
- **Negative weights:** Corrected with `abs()`; if some are sentinel values, imputation should be revisited.